New Idea For Requirement

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1. Make Cabinet Can Contain more Item

Why?

- Cabinet will have more variety of thing people can borrow and more amount as well

How?

- Using Bigger Cabinet
- Using Fish Eye Lens to make minimum distance from Camera to item shorter so we can add more Floor in Cabinet
- Make a Floor to be ลิ้นชัก so it easily to pick up item even if in 1 cabinet are pack with a lot of floor

2. Have AI Chatbot in student website

Why?

- Some IOT Sensor names as a code such as DHT11 which is a temperature sensor. It can make a student who wants any temperature sensor suffer from not knowing what sensor he needs to find. So I think if we have an AI Chatbot. it will help student find what they need easier

How?

- Let Yanin and Syril Handle this

3. Increase AI Model accuracy for identify item from 70% to 90%

Why?

- Right now my AI has accuracy around 70% which sucks. if we use it in real production a lot of people will mark as borrowing something that they do not borrow.

How?

- Let Yanin and Syril Handle this

4. Find the way to Make Register Item for Staff Easiest as Possible

Why?

- The reason electrical engineers don't use their own website anymore is because in real situations they have 6000 items but each item takes a lot of time to register so they give up at 600 items. we need to solve this problem

How?

- Read Paper about this and hope we will find solution

Existing Requirements and Current Capabilities

Smart Inventory System

The Smart Inventory System was originally developed to address three main problems: limited visibility of available equipment, missing or untraceable items, and the administrative workload caused by manual borrowing records. The project combines a web application, backend services, IoT cabinet hardware, NFC authentication, and an AI-based equipment detection system.

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1. Existing Functional Requirements

FR-1: User Authentication via NFC

The system shall allow registered users to authenticate themselves by scanning their NFC cards. Once the card is successfully verified, the system shall unlock the cabinet and create an active cabinet-access session.

The user registration process shall also support linking and unlinking an NFC card to a user account.

FR-2: Automated Equipment Borrowing

The system shall allow users to borrow equipment by performing the following steps:

1. Scan a registered NFC card.
2. Open the cabinet.
3. Remove the required equipment.
4. Close the cabinet.

After the cabinet is closed, the system shall use the built-in camera and AI system to identify which items were removed. The borrowing transaction shall then be recorded automatically without requiring the user to manually select the equipment on a screen.

The borrowing record shall include:

- User information
- Borrowed item
- Borrowing date and time
- Return due date
- Current borrowing status

FR-3: Automated Equipment Return

The system shall allow users to return equipment by opening the cabinet, placing the equipment inside, and closing the cabinet.

After the cabinet is closed, the system shall capture an image and use AI to determine which items were returned. When the return is successfully verified, the system shall update the item status to available and record the return date and time.

FR-4: View Available Equipment

The system shall allow students, registered users, and guests to view the list of available equipment through the web application.

Guest users shall be able to view equipment availability without registering or logging in.

The displayed information may include:

- Equipment name
- Equipment image
- Availability status
- Storage or cabinet location

FR-5: View Borrowing History

The system shall allow authenticated users to view their complete borrowing and returning history through the web interface.

The history shall include:

- Equipment details
- Borrowing date and time
- Return date and time
- Return due date
- Transaction status

FR-6: Due-Date Email Notifications

The system shall send email notifications to users when:

- The equipment return due date is approaching.
- The borrowed equipment has passed its return due date.

These notifications are intended to reduce the need for instructors or administrators to manually follow up with students.

FR-7: Real-Time Transaction Confirmation

After the cabinet is closed and the AI analysis is completed, the system shall display a real-time confirmation showing which items were borrowed or returned.

This confirmation allows the user to verify that the system recorded the transaction correctly.

FR-8: Damaged Equipment Reporting

The system shall allow users to report damaged or broken equipment through the web application.

A reported item shall require administrator review before the return process is considered complete. The administrator shall be able to approve or reject the damage report.

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2. Existing User Roles

2.1 Student or Normal User

A student or normal user can currently perform the following actions:

- Register a user account
- Link or unlink an NFC card
- View available equipment
- View currently borrowed equipment
- Borrow equipment through the smart cabinet
- Return equipment through the smart cabinet
- View personal borrowing and returning history
- Receive real-time transaction confirmation
- Submit damaged-equipment reports
- Receive due-date and overdue email notifications

2.2 Administrator

An administrator can perform all normal-user functions and additional administrative functions, including:

- Register new equipment
- Upload equipment reference images
- Manage equipment information
- Manage users
- View and manage borrowing records
- View current equipment holders
- View cabinet-access sessions
- View activity and audit logs
- View AI-captured images
- Review damaged-equipment reports
- Approve or reject damaged-equipment reports

3. Existing System Capabilities

3.1 Web Application

The project includes a web application developed with Next.js, TypeScript, and Tailwind CSS.

Student Portal

The student portal currently supports:

- Viewing real-time equipment availability without logging in
- Viewing borrowed equipment
- Viewing borrowing and returning history
- Viewing transaction confirmation after a cabinet session
- Submitting damaged-equipment reports

Administrator Dashboard

The administrator dashboard currently supports:

- Viewing the live inventory status
- Identifying which user currently holds each item
- Managing equipment records
- Managing borrowing transactions
- Managing users
- Viewing activity and cabinet-session logs
- Reviewing AI-captured images
- Managing damaged-equipment reports
- Registering new equipment with reference images

3.2 Backend System

The backend is developed using FastAPI and follows a modular, monolith-inspired architecture.

The backend currently includes the following services.

Authentication Service

The Authentication Service is responsible for:

- NFC card verification
- User authentication
- JWT token generation

- User-session management

Inventory Service

The Inventory Service is responsible for:

- Creating, viewing, updating, and deleting equipment records
- Managing equipment availability
- Processing borrowing transactions
- Processing return transactions
- Managing damaged-equipment reports

Notification Service

The Notification Service uses a background due-date checker to:

- Identify equipment approaching its due date
- Identify overdue equipment
- Send email notifications through Gmail SMTP

Cabinet Session Service

The Cabinet Session Service records each cabinet-access session, including:

- Authenticated user
- Cabinet opening time
- Cabinet closing time
- Active session status
- Equipment changes detected during the session

MQTT Bridge

The MQTT Bridge provides communication between the cabinet hardware and the backend.

It can receive hardware events such as:

- NFC card scans
- Cabinet door opening
- Cabinet door closing
- Sensor-status updates

It can also send commands back to the ESP32, such as cabinet-unlock commands.

3.3 Smart Cabinet and IoT Hardware

The existing cabinet prototype includes the following hardware components:

- ESP32

- ESP32-CAM
- NFC reader
- Solenoid lock
- Relay modules
- Magnetic contact sensor
- Buzzer module
- Power-regulation modules

The cabinet can currently:

- Read NFC cards
- Send NFC card information to the backend
- Unlock the cabinet after successful authentication
- Detect whether the cabinet door is open or closed
- Capture images of the equipment inside the cabinet
- Send hardware events through MQTT
- Produce sound notifications through the buzzer

3.4 AI-Based Equipment Detection

The project includes an AI system for detecting equipment that has been borrowed or returned.

Before-and-After Image Comparison

The system compares two cabinet images:

- **Baseline image:** The image captured after the previous cabinet session
- **Current image:** The image captured after the current cabinet session

The system analyses the differences between the images.

- An item that disappears may be classified as borrowed.
- An item that appears may be classified as returned.

Object Detection

The system uses a YOLO model to:

- Detect equipment inside the cabinet
- Generate bounding boxes around detected objects
- Crop individual equipment images
- Prepare cropped images for identification

Item Identification

Each cropped equipment image is converted into an embedding vector.

The embedding is compared with stored reference embeddings using cosine similarity. The system identifies the equipment when the similarity score meets the defined threshold.

This design allows new equipment to be registered by uploading reference images without retraining the complete object-detection model.

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4. Existing Data and Storage Capabilities

The current system uses:

- PostgreSQL for the production database
- SQLite for the development environment
- SQLAlchemy as the Object-Relational Mapping framework
- S3-compatible storage for equipment and AI-captured images

The system stores information related to:

- Users
- NFC cards
- Equipment
- Equipment availability
- Borrowing transactions
- Return transactions
- Cabinet-access sessions
- Activity logs
- Damaged-equipment reports
- Equipment reference images
- AI-captured cabinet images

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5. Existing Testing Capabilities

5.1 Backend Testing

The backend includes:

- Nine unit-test files using Pytest
- Eight end-to-end system-test suites using Robot Framework

The existing tests cover major backend functions, including:

- Authentication
- User management
- Equipment management
- Borrowing transactions

- Return transactions
- Damage reports
- Activity logs

One existing test verifies that the system rejects an attempt to borrow equipment that is already checked out.

Another test simulates a complete borrowing and returning process and verifies that the equipment status returns to available after a successful return.

5.2 Frontend Testing

The frontend uses Jest and React Testing Library.

The tests cover:

- Component rendering
- State changes
- User interactions
- Form validation
- API calls
- Administrator-page functions
- Search and filtering
- Confirmation prompts

For example, the damaged-equipment report form is tested to ensure that incomplete information cannot be submitted.